

JMM Abstract

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Normalizing flows construct mapping from a complex target distribution to a standard normal using a sequence of invertible transformations. Given that there can be infinitely many transformations between two distributions, in this work we take a perspective of optimal transport to find the map by enforcing convexity to align with the solution of Brenier’s theorem, which guarantees that the map is the gradient of a convex potential. We seek the map given only samples from the source and target distributions. In this work, we explore two parameterizations of the convex potential. The first uses convex radial basis functions (RBFs), where we introduced two novel modifications to the literature: (i) evaluating gradients at the original source points rather than the transported particles, and (ii) enforcing convexity at each step through a linear program derived from Farkas’ Lemma. These additions significantly improved theoretical consistency of the resulting maps. The second employs Input Convex Neural Networks (ICNNs), which guarantee convexity by construction through architectural constraints on the weights and activations. Both approaches preserve the convexity structure required by optimal transport, ensuring that the learned maps remain compatible with Brenier’s theorem. We implemented the ICNN-based approach on synthetic datasets, including banana-shaped and ring-shaped distributions, and demonstrate that ICNNs successfully capture complex nonlinear transformations while maintaining convexity. These results highlight the promise of convexity-preserving normalizing flows as a bridge between rigorous transport theory and practical generative modeling.